

Self Project

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# Vehicle-Scale LLMs: Low-Rank Residuals + 4-Bit Quantization for In-Vehicle AI

A project walkthrough of DAC+Q4-ITS, a minimal reproducible starter for INT4 quantization and low-rank adapter compensation for vehicle-scale LLM inference.

## Project Snapshot

### DATE

2025-08-22

### CATEGORY

Self Project

### SHORT DESCRIPTION

A project walkthrough of DAC+Q4-ITS, a minimal reproducible starter for INT4 quantization and low-rank adapter compensation for vehicle-scale LLM inference.

### TAGS / TECH STACK

LLM Inference

INT4 Quantization

Low-Rank Adapters

Edge AI

Automotive AI

PyTorch

## Project Links

[GitHub Repository](#)

## Full Project Content

A compact starter for memory-efficient, vehicle-oriented LLM inference using **INT4 quantization** and **low-rank adapter compensation**.

**GitHub Repository:** [dranubhaparashar/Vehicle-Scale-LLMs-Integrating-Low-Rank-Residuals-and-4-Bit-Quantization-for-In-Vehicle-AI](https://github.com/dranubhaparashar/Vehicle-Scale-LLMs-Integrating-Low-Rank-Residuals-and-4-Bit-Quantization-for-In-Vehicle-AI)

This post documents a compact, reproducible starter for **vehicle-scale LLM inference**. The repository presents **DAC+Q4-ITS** as a minimal pipeline that demonstrates a complete flow from corpus preparation to export, INT4 quantization, eigenspace computation, adapter injection, inference, and evaluation.

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## Vision

Large language models can be useful in in-vehicle AI systems, but deployment is difficult when memory and compute budgets are tight. This project explores a practical approach: combine **4-bit quantization** with **low-rank residual compensation** so the model remains lightweight while recovering part of the quality lost through aggressive compression.

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## Why This Matters

**Note** The repository is explicitly framed as a **minimal reproducible starter** for an **INT4 + low-rank adapter compensation pipeline**.

**Important** The README describes an end-to-end flow: **corpus** → **export** → **quantize (INT4)** → **eigenspace (SVD)** → **build adapters (U, D)** → **inject** → **inference** → **evaluation**.

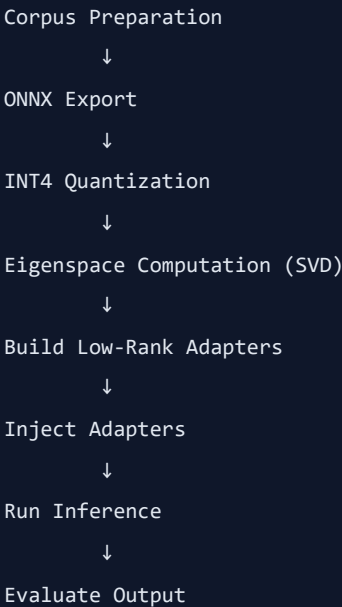
**Tip** The starter is designed to prove the flow on a toy transformer first, then be swapped with a LLaMA-derived model later.

**Warning** The current public repo is a starter implementation, not a full production deployment kit.

**Caution** Claims about real-world automotive performance should be validated with a real

model, real deployment stack, and benchmark data.

## What This Starter Does



The repository README describes this exact flow and positions it as a proof-of-pipeline for memory-efficient inference.

## Project Attributes

| Attribute         | Description                                                                                                    |
|-------------------|----------------------------------------------------------------------------------------------------------------|
| problem-statement | Full-precision LLM inference is too heavy for constrained in-vehicle environments.                             |
| primary-objective | Demonstrate a memory-efficient inference pipeline using INT4 quantization plus low-rank residual compensation. |
| core-technologies | PyTorch, ONNX export, INT4 quantization, SVD/eigenspace methods, low-rank adapters.                            |
| target-setting    | In-vehicle AI and other edge-like environments where memory and compute are limited.                           |
| key-mechanism     | Compressed forward pass described by the repo as $\text{dequantized}(\text{INT4 matmul}) + U(Dx)$ .            |
| current-scope     | A minimal reproducible starter built around a toy transformer.                                                 |

| Attribute        | Description                                                                            |
|------------------|----------------------------------------------------------------------------------------|
| future-direction | Swap the toy model with a LLaMA-derived model and wire TensorRT for Jetson when ready. |

## Repository Structure

The repository currently includes dedicated folders for configs, data, Docker, docs, environment setup, scripts, templates, tests, and source code under `src/dac_q4_its`.

```
Vehicle-Scale-LLMs-Integrating-Low-Rank-Residuals-and-4-Bit-Quantization-for-In-Vehicle-AI/  
├─ configs/  
├─ data/  
├─ docker/  
├─ docs/  
├─ env/  
├─ guidelines/  
├─ scripts/  
├─ src/  
│   └─ dac_q4_its/  
├─ templates/  
├─ tests/  
├─ Makefile  
├─ README.md  
└─ pyproject.toml
```

## Quickstart

The README provides this starter flow:

```
python -m venv .venv && source .venv/bin/activate  
pip install -r env/requirements.txt  
python scripts/01_prepare_calibration.py  
python scripts/02_export_onnx.py  
python scripts/03_quantize_int4.py  
python scripts/04_compute_eigenspaces.py  
python scripts/05_inject_adapters.py  
python scripts/07_run_inference.py --prompt "avoid tolls and reach airport by 6pm"  
python scripts/08_eval_nln.py
```

This sequence shows the intended end-to-end execution path from calibration preparation to inference and evaluation.

## What the Starter Gives

According to the README, the starter includes:

- Working INT4 quantization with per-row scales
  - Rank-8 adapters computed from calibration activations
  - A compressed forward pass based on dequantized INT4 matmul plus low-rank residual correction
  - Simple EM / SOFT-F1 metrics demo
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## Core Compression Idea

```
def compressed_forward(x, w_int4, scales, U, D):  
    base = dequantized_int4_matmul(x, w_int4, scales)  
    correction = U @ (D @ x)  
    return base + correction
```

The exact README wording summarizes the compressed forward path as:

```
dequantized(INT4 matmul) + U(Dx)
```

That is the central systems idea of this starter.

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## Why This Design Is Interesting

- It targets memory efficiency directly through **INT4 quantization**
  - It adds a **low-rank correction path** instead of fully undoing compression
  - It keeps the pipeline modular, with separate scripts for export, quantization, eigenspace computation, adapter injection, inference, and evaluation
  - It is positioned as a bridge from a toy transformer to a real LLaMA-derived deployment path
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## Real-Model Upgrade Path

The README suggests the following path to move beyond the toy transformer:

- Replace `src/dac_q4_its/modeling/loader.py` to load real HF / LLaMA weights
- Populate `scripts/02_export_onnx.py` to export real projections
- Replace toy heads with real Q / K / V / FFN matrices
- Keep the rest of the pipeline structure intact

This is useful because it shows that the starter is intentionally modular rather than being locked to the toy model.

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## Current Limitations

- Public repo is still a starter, not a fully benchmarked release
- README is concise and focused on flow, not exhaustive evaluation
- Real deployment details for Jetson / TensorRT are not yet fully wired in the public starter
- The repo currently proves pipeline structure more than production-grade validation

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## What I Would Add Next

- Benchmark comparisons against FP16 / INT8 / plain INT4 baselines
- Memory usage charts for embedded automotive targets
- End-to-end latency numbers on Jetson-class hardware
- Accuracy trade-off analysis as rank changes
- A worked example using a real LLaMA-derived checkpoint

These additions would make the repo stronger as both a research artifact and an engineering deployment guide.

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## Key Innovation

**Important** The main innovation is the combination of **INT4 quantization** with a **low-rank residual compensation path** in a pipeline designed for constrained, vehicle-oriented inference settings.

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## Conclusion

This repository is a strong starter for exploring **vehicle-scale LLM inference** under tight memory budgets. Its value is not only in compression, but in the fact that it demonstrates a full reproducible flow: prepare, export, quantize, compensate, inject, infer, and evaluate.

From full-precision overhead to **vehicle-ready efficient inference** — that is the direction this project is pushing toward.